

ClassMic Business Case

B2 Investment decision for the course prototype

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Recommendation

GO for a limited course prototype using existing resources. Build the browser workflow to test the ClassMic idea and complete the course learning objectives. Keep any funded classroom pilot at REVISE until local need, audio performance, support effort and a commercial comparison are established.

The decision requested is permission to investigate, prototype and evaluate. The preliminary plan allows 100 student hours and assumes no new cash spending. Use these as the planning baseline and check actual capacity before each sprint.

Opportunity and expected benefit

ClassMic aims to let the room hear student questions without waiting for a microphone to reach the speaker. Students join a temporary browser session, request a turn and transmit audio only after instructor approval. The instructor can mute a speaker or end the session.

Expected value is easier audibility and less interruption during discussion. The course also gains a concrete way to study requirements, project control and iterative delivery. Improved grades, increased participation and institutional savings remain hypotheses. The present evidence cannot support a monetary return calculation.

Evidence and its limits

Evidence	What it supports	What remains unknown
In-class SMART note	The phone microphone concept and intended outcomes.	How often students miss questions and whether they want this workflow.
Existing ClassMic drafts	A starting scope and list of risks.	Approval, available resources and measured feasibility.
Biamp product information [1]	A commercial product offers the same core microphone and moderator workflow.	Local price, room fit and comparison with our prototype.
Browser media specification [3]	Browser microphone capture is technically supported with permission controls.	Performance and compatibility on the devices and network we will use.

If we do nothing

Continue current classroom practice. This avoids development and maintenance work, but leaves any audibility problem unchanged. We should observe the current process before claiming that ClassMic improves it.

Alternatives appraisal

Compare the same classroom objective across three options. The judgments below are preliminary. Unknown costs and untested adoption are left unresolved rather than converted into a precise score.

Criterion	Improve current process	Buy or adapt	Custom ClassMic prototype
Strategic fit	Support discussion by repeating questions or using an existing microphone.	Existing audience microphone workflow may fit the classroom.	Focused test of browser access and instructor control.
Expected benefit	May solve the problem with a small change in teaching practice.	Phone speaking and moderation are established product capabilities [1].	Benefit must be demonstrated against the current workaround.
Cost and value	No new platform. Staff time and equipment availability still matter.	Room hardware and installation require a quote [2]. Total cost unknown.	Proposed 100 student hours. No incremental cash assumed. Maintenance is unresolved.
Feasibility	Lowest technical complexity if the existing workaround is adequate.	Needs procurement, AV integration and a product trial.	Browser, network, feedback and permission behavior need testing.
Schedule	Can begin after agreement on the classroom routine.	Delivery and installation dates require a supplier response.	Two weekly sprints are planned. Audio is the main schedule risk.
Security and privacy	No new microphone software or data flow.	Review actual configuration, data handling and support access.	Control data capture and avoid saved speech. The team owns testing.
Operational support	Uses current teaching and AV support arrangements.	Supplier and institution divide support duties. Terms are unknown.	Two students support the prototype. Production support is outside scope.
Stakeholder adoption	Familiar process, but repeated questions may remain disruptive.	Test the attendee and moderator workflow with real users.	No-install browser access is a hypothesis to validate with students.

Reason for the recommendation

A custom prototype fits the course requirement to build and evaluate selected user stories. That educational reason does not establish that a university should build its own microphone service. A commercial trial and an improved current process remain credible options for a later institutional decision.

Proposed delivery and resources

Stage	Planned dates	Decision or output
Initiation and planning	10–19 September	Review A2 B2 and C1. Complete requirements and D plan before the first sprint.
Sprint 1	20–26 September	Temporary sessions, permission flow, queue and instructor controls. Review the working flow.
Sprint 2	27 September–3 October	Integrate audio, test misuse and recovery, collect volunteer feedback and report results.
Closing	4–9 October	Assess delivery against the charter and prepare the final recommendation.

The LMS deadlines are D on 22 September, E on 4 October and F on 10 October, each at 00:00 as displayed. Schedule D review for 19 September so that both full weekly sprints fit. The E upload area opens on 26 September; this controls uploading and does not state a restriction on earlier construction. If the D review is late, resolve the resulting schedule conflict before changing the sprint lengths.

Preliminary cost and effort

Work	Temiko hours	Nurtore hours	Total hours
Initiation and D planning	18	22	40
Sprint 1	18	6	24
Sprint 2	14	10	24
Closing and presentation	4	8	12
Total hours	54	46	100

Cash plan: AED 0 of new spending, conditional on reusing available phones, laptops, speakers and permitted tools. Student effort is in kind, not free of opportunity cost. No labor rate or supplier quote is available, so this case does not invent a cash equivalent. Any paid hosting, equipment or account upgrade requires a revised estimate and approval before commitment.

Feasibility conditions

Confirm access to at least two suitable phones, a laptop and a speaker. Test microphone permission and browser behavior early. Use volunteer tests in a controlled setting. Room access, advisers and participants are not yet committed.

The selected prototype target is a 95th percentile one-way audio delay of no more than 150 ms across 30 timing trials. This is a planning target, not a tested result. Also assess intelligibility and sustained feedback; a delay number alone does not establish usable room sound. A commercial comparison gives context [2].

Decision gates and principal risks

Gate	Evidence required	Decision
Before Sprint 1	Team and instructor review, five-story scope, resources, test plan and proposed schedule.	GO with the prototype only if the plan is feasible. Otherwise revise scope or dates.
After Sprint 1	Session and queue demo, microphone feasibility check, feedback and updated backlog.	Continue core audio integration. Drop optional work if capacity or compatibility is weak.
End of Sprint 2	All Must checks. 9/10 setups within 2 minutes, 27/30 speaking trials within 3 seconds, and at least 75% positive feedback from at least 5 reviewers. No critical defect.	Complete the course evaluation. Report unmet targets and test limitations.
Any future live pilot	Local need, acceptable audio, named support owner, privacy and access review, supplier comparison and funded estimate.	REVISE until evidence is sufficient. Seek a separate authorization.

Risks that could change the decision

Feedback or network delay may make the audio unsuitable. Browser differences may prevent some students from using it. Weak instructor controls could permit unwanted transmission. Limited student time or unavailable reviewers could prevent a useful evaluation. Stop an affected test if control or privacy fails, preserve a physical or instructor-repeat alternative, and narrow the prototype before adding resources.

Investment conclusion

The present evidence supports a bounded learning project. It does not justify a funded institutional rollout or a claim that custom development is the best long-term purchase decision. At closeout, recommend further testing only if the prototype demonstrates useful audibility, workable instructor control and a realistic support path.

Items requiring confirmation

Confirm the revised A2 and B2, team capacity, early D approval, test equipment and volunteers. Use the selected 150 ms delay target and the supplied test-data protocol. Record the instructor's decision. Do not infer approval from an LMS upload.

Sources

[1] [Biamp Crowd Mics product overview](#)

[2] [Biamp Crowd Mics FAQ, including quote requirements and audio latency](#)

[3] [W3C Media Capture and Streams specification](#)

External sources reviewed 10 September 2026. Local inputs: the ClassMic in-class SMART note and existing charter and plan. Course requirements: the LMS B, C, D, E and F pages and D Phase Labs attachment. No stakeholder interviews, supplier quotation or pilot results were supplied.